Exp 25 MATLAB CODE

```
clc;
clear;
close all;

% Generate Random Input Bits
data = randi([0 1],1,20);

% Convolutional Encoder Parameters
trellis = poly2trellis(3,[7 5]);

% Convolutional Encoding
encoded_data = convenc(data,trellis);

% Display Input and Encoded Bits
disp('Input Bit Sequence:');
disp(data);

disp('Encoded Bit Sequence:');
disp(encoded_data);

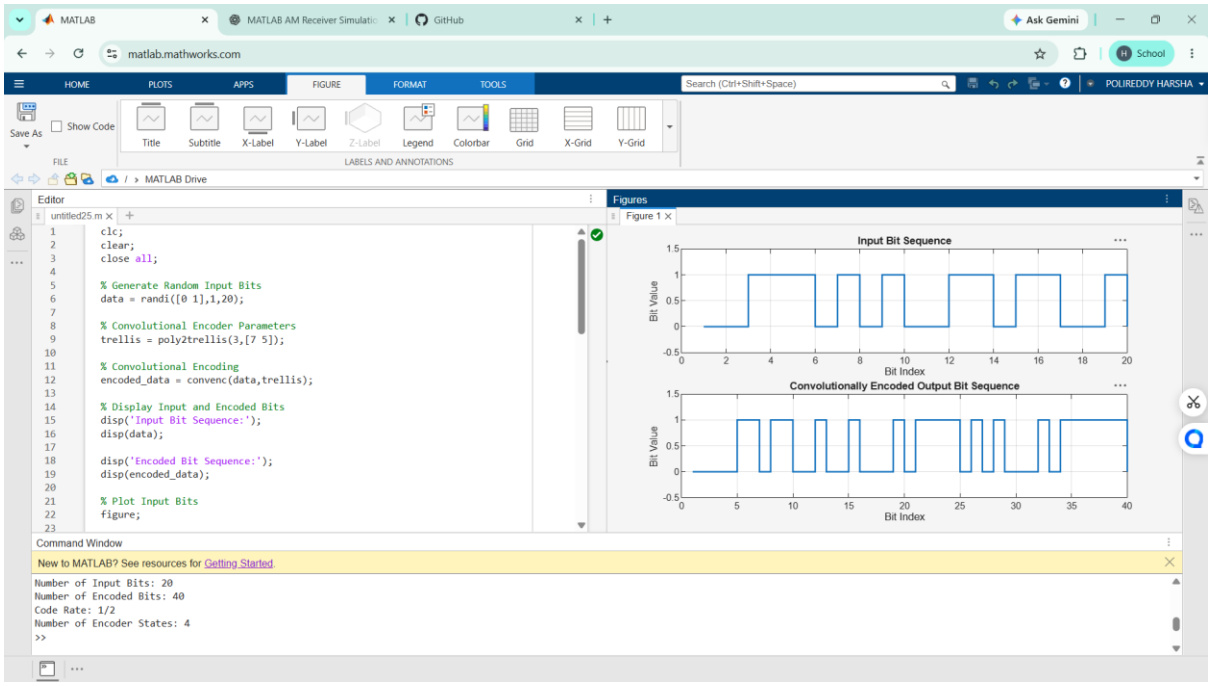
% Plot Input Bits
figure;

subplot(2,1,1);
stairs(data,'LineWidth',1.5);
title('Input Bit Sequence');
xlabel('Bit Index');
```

```
ylabel('Bit Value');
ylim([-0.5 1.5]);
grid on;

% Plot Encoded Bits
subplot(2,1,2);
stairs(encoded_data,'LineWidth',1.5);
title('Convolutionally Encoded Output Bit Sequence');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.5 1.5]);
grid on;

% Analysis
fprintf('Number of Input Bits: %d\n',length(data));
fprintf('Number of Encoded Bits: %d\n',length(encoded_data));
fprintf('Code Rate: 1/2\n');
fprintf('Number of Encoder States: %d\n',trellis.numStates);
```


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Y-Grid
LABELS AND ANNOTATIONS
/ › MATLAB Drive
Editor
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1 clc;
2 clear;
3 close all;
4
5 % Generate Random Input Bits
6 data = randi([0 1],1,20);
7
8 % Convolutional Encoder Parameters
9 trellis = poly2trellis(3,[7 5]);
10
11 % Convolutional Encoding
12 encoded_data = convenc(data,trellis);
13
14 % Display Input and Encoded Bits
15 disp('Input Bit Sequence:');
16 disp(data);
17
18 disp('Encoded Bit Sequence:');
19 disp(encoded_data);
20
21 % Plot Input Bits
22 figure;
23
Figures
Figure 1
Input Bit Sequence
Bit Value
Bit Index
Convolutionally Encoded Output Bit Sequence
Bit Value
Bit Index
Command Window
New to MATLAB? See resources for Getting Started.
Number of Input Bits: 20
Number of Encoded Bits: 40
Code Rate: 1/2
Number of Encoder States: 4
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